Isaac ROS Object Detection

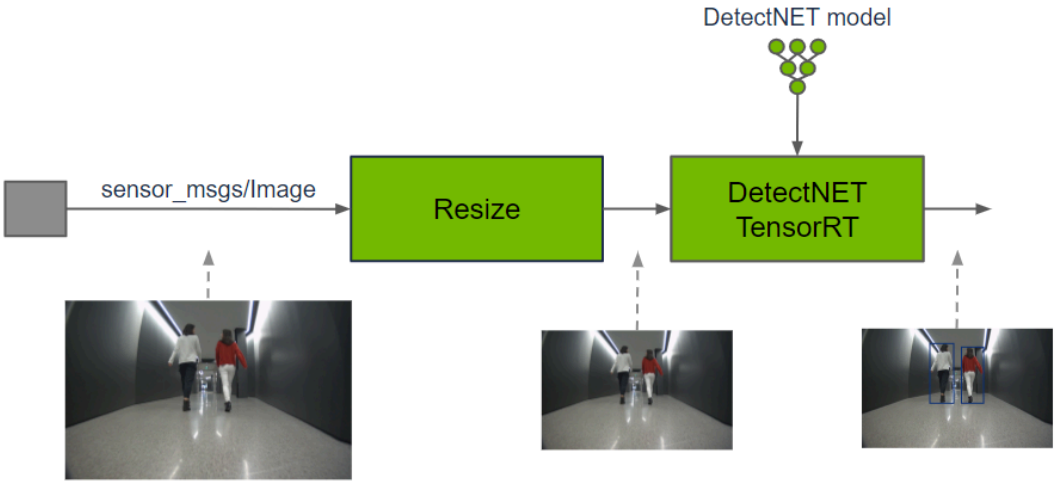
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Isaac ROS Object Detection official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_object_detection/index.html

Overview

Isaac ROS provides GPU-accelerated detection models such as RT-DETR, DetectNet, and YOLOv8, which can predict the target class and 2D bounding box from the input image.



Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

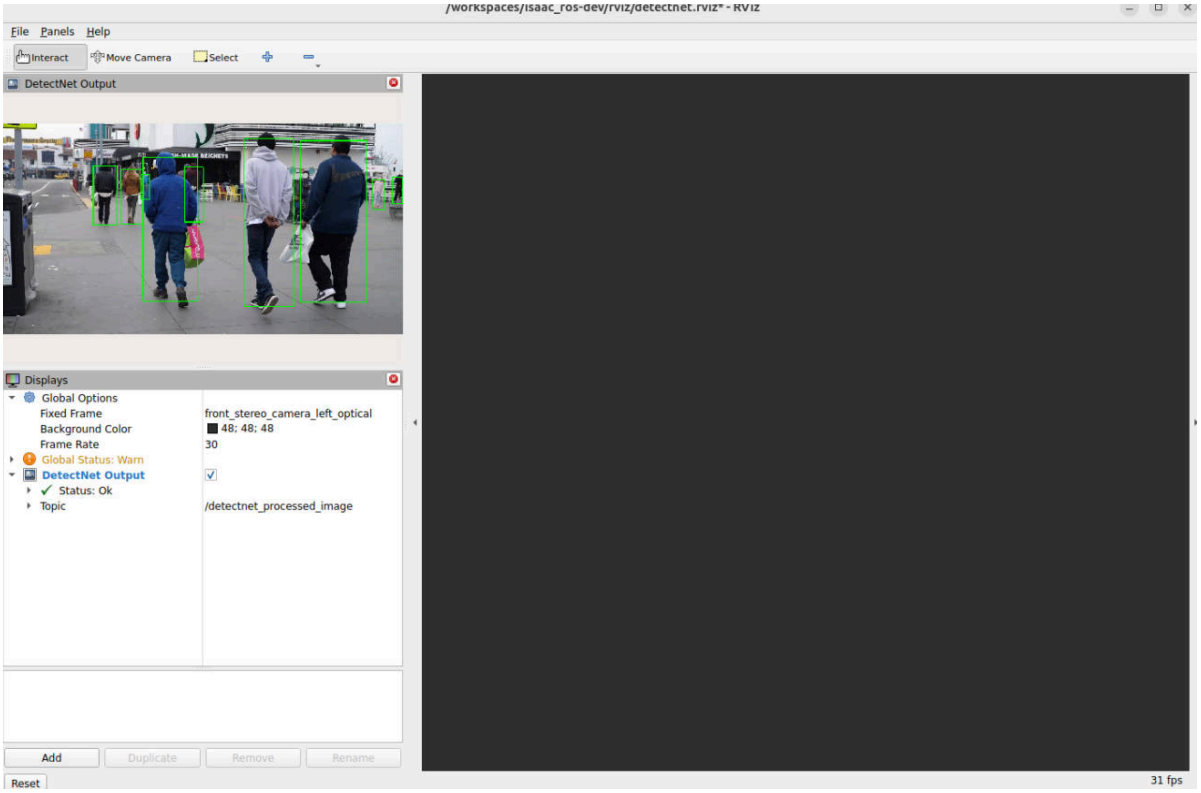
Terminal A: Start DetectNet, the synchronized visualization node, and RViz:

```
ros2 launch /workspaces/isaac_ros-dev/launch/detectnet_all.launch.py
```

The RViz configuration is `/workspaces/isaac_ros-dev/rviz/detectnet.rviz`; view `/detectnet_processed_image`.

Terminal B: Play the detection bag in a loop at 0.25× speed:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_detectnet/rosbags/detectnet_rosbag \
  --loop --rate 0.25 \
  --remap /image:=/image_rect /camera_info:=/camera_info_rect
```



Terminal C: Check the detection and drawn image outputs:

```
timeout 8 ros2 topic hz /detectnet/detections
timeout 8 ros2 topic hz /detectnet_processed_image
ros2 topic info -v /detectnet_processed_image
```